

UNIVERSITI TUNKU ABDUL RAHMAN

ACADEMIC YEAR 2023/2024

SEPTEMBER EXAMINATION

UCCD1013 ANALYSIS AND DESIGN OF INFORMATION SYSTEMS

MONDAY, 9 OCTOBER 2023

TIME : 2.00 PM – 4.00 PM (2 HOURS)

BACHELOR OF COMPUTER SCIENCE (HONOURS)
BACHELOR OF INFORMATION SYSTEMS (HONOURS)
BUSINESS INFORMATION SYSTEMS
BACHELOR OF INFORMATION SYSTEMS (HONOURS)
DIGITAL ECONOMY TECHNOLOGY
BACHELOR OF INFORMATION TECHNOLOGY (HONOURS)
COMPUTER ENGINEERING

Instructions to Candidates:

This question paper consists of **THREE (3)** questions in **Section A** and **TWO (2)** questions in **Section B**.

Answer **ALL** questions in **Section A** and **CHOOSE ANY ONE (1)** question in **Section B**.
Each question carries 25 marks each.

Note: Should you answer more than ONE (1) question in section B, marks will only be awarded to the FIRST question attempted.

UCCD1013 ANALYSIS AND DESIGN OF INFORMATION SYSTEMS**Section A (Answer ALL Questions)**

- Q1. (a) Table 1.0 shows the sequences of activities in XYZ's project for software development.

Activity	Expected Time	Predecessor
1. Requirement Collection	5	-
2. Requirement Analysis	3	1
3. Prototype & Analysis	4	1,2
4. Programming & Testing	20	2,3
5. Installation	6	4

Table 1.0: Sequence of activities in XYZ's project.

- (i) Calculate the Early Finish (EF) values for the activities listed in Table 1.0. (5 marks)
- (ii) Calculate the Late Finish (LF) values for the activities listed in Table 1.0. (5 marks)
- (b) System requirements determination is divided into traditional and contemporary methods. State **THREE (3)** contemporary methods for determining system requirements. (6 marks)
- (c) Given Table 2.0 on the worksheet for ABC Software Firm Project, assuming the monetary benefit of ABC Software Firm starts at the second year onwards, which is RM5600 in the second year, RM8200 in the third & fourth year. While the one-time costs are RM3000 and the recurring costs are RM2300 for the first, second, third, and fourth year, respectively.

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	Year 0 (RM)	Year 1 (RM)	Year 2 (RM)	Year 3 (RM)	Year 4 (RM)	Total (RM)
Net Economic Benefit	0	0	5600	8200	8200	
Discount rate (10%)	1.0000	0.9091	0.8265	0.7513	0.6830	
PV of Benefits	0	0	4628.40	6160.66	5600.60	
NPV of all Benefits	0	0	4628.40	10789.06	(a)	(a)
One time cost	(3000)					
Recurring cost	0	(2300)	(2300)	(2300)	(2300)	
Discount rate (10%)	1.0000	0.9091	0.8265	0.7513	0.6830	
PV of Recurring Costs	0	(2090.93)	(1900.95)	(1727.99)	(1570.90)	
NPV of all Costs	(3000)	(5090.93)	(6991.88)	(8719.87)	(b)	(b)
Overall NPV						(c)
ROI (%)						(d)

Table 2.0: Economic Feasibility Study for ABC Software Firm Project.

- (i) Workout the economic feasibility study by using the concept of Time Value of Money (TVM) by listing the values at (a) – (d) columns shown in Table 2.0. **Show your answers in two decimal places.**
(5 marks)
- (ii) In your opinion, from the calculated values of Overall Net Present Value (NPV) and Return on Investment (ROI) in Q1(c)(i), state if the ABC Software Firm Project is generating profit or otherwise. Justify your answer.
(4 marks)
[Total : 25 marks]

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- Q2. (a) Acceptance testing is performed by the real users to determine if the system meets their requirements. Alpha testing and Beta testing are the two types of acceptance testing.
- (i) Discuss the differences between Alpha testing and Beta testing. (4 marks)
 - (ii) Identify and explain **TWO (2)** types of Alpha testing. (8 marks)
- (b) Installation is the organizational process of changing over from the current information system to a new one. Parallel installation strategy is one type of installation strategies. Discuss the parallel installation strategy. (6 marks)
- (c) Indicate **ONE (1)** information system that you had designed in your assignment. As a developer of the system, explain how single-location installation strategy shall be beneficial for the installation process of the system. (7 marks)
- [Total : 25 marks]

- Q3. (a) System maintenance encompasses of processes for improving a system for a smooth operation preventing it from bugs or errors. There are four types of system maintenances.
- (i) Perfective maintenance is one type of system maintenances. Discuss perfective maintenance. (8 marks)
 - (ii) XYZ corporation has been using an attendance system that relies on recording employees' attendance through scanning their staff identification (ID) card. However, the corporation has been encountering challenges in using the mentioned system, efficiently. This is mainly because of loss or damage of the staff ID card. Employees misplace or damage their ID card, making it difficult to record their attendance, accurately via the system. They may need to seek for replacements of ID card, which shall cause delays and administrative overheads. As a system maintenance personnel for the XYZ corporation, you have decided to perform adaptive maintenance to the attendance system. Justify why you must perform adaptive maintenance to the mentioned system. (8 marks)
- (b) In an organization, maintenance personnel are the developers who are responsible in the maintenance of an information system.
- (i) Determine the **THREE (3)** types of maintenance personnel approach found in an organizational structure. (3 marks)

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- (ii) From a maintenance personnel's perception, discuss **ONE (1)** of the maintenance personnel approach from the answers in Q3(b)(i) which is bad practiced in an organization, in terms of cost and time effectiveness. (6 marks)

[Total : 25 marks]

Section B (Choose any ONE question)

- Q4. (a) SEM electronics is a retail store that sells electronic devices and accessories. The store management wants to implement a sales and inventory management system to efficiently handle their operations. In the system design, customers shall submit a purchase order form for verification. The verified order details are processed to be sent as a dispatch order to the warehouse and an invoice shall be generated to the customer from them. Moreover, these verified order details shall be recorded as sales in customer's sales file and as stock depletion amount in stock file. These records of sales and stock depletion amount shall be generated as a report to the store manager.

Given the mentioned scenario, prepare the items listed below to form a data flow diagram (DFD).

- (i) Determine **THREE (3)** sources/sinks and **TWO (2)** data stores for the given scenario. (5 marks)
- (ii) Determine **FOUR (4)** processes for the given scenario. (4 marks)
- (iii) Draw a data flow diagram (DFD) level 0 using the listed sources/sinks and data stores from Q4(a)(i) and the processes identified in Q4(a)(ii) in correct sequence. (10 marks)
- (b) The selection of hardware device(s) for system interfaces interactions must be made during logical design as different interfaces require different device(s). Identify **THREE (3)** types of hardware devices used for system interfaces interactions. (6 marks)

[Total : 25 marks]

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- Q5. (a) Consider a university's course registration system. A student can register for one or all courses structured in the programme structure of an academic semester using the system. While a course should have a minimum of ten students registered to be continued for the semester. Once the course(s) registration is confirmed, the university shall prepare a bill for the student for enrollment to the semester. If the student wishes to add or drop any registered course(s) after the bill is generated, he/she has to do it via the system. New bill shall be generated to the student for the semester's enrollment by the university after the add or drop of registered course(s). The student has to pay the bill by week seven of the academic semester for examination seating.

Given the mentioned scenario, prepare the items listed below to perform normalization using third normal form (3NF).

- (i) Determine **THREE (3)** relations with **THREE (3)** attributes for the given scenario. (4 marks)
 - (ii) Determine **First Normal Form (1NF)** for the relations identified in Q5(a)(i). (3 marks)
 - (iii) Determine **Second Normal Form (2NF)** for the 1NF relations identified in Q5(a)(ii). (3 marks)
 - (iv) Determine **Third Normal Form (3NF)** for the 2NF relations identified in Q5(a)(iii). (3 marks)
 - (v) Determine if **associative entity(s)** exist for the 3NF relations identified in Q5(a)(iv). Justify your answer. (6 marks)
- (b) Competitive strategy is a method for organization to attempt to satisfy its mission and objectives. State **THREE (3)** types of competitive strategy. (6 marks)
- [Total : 25 marks]
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